

Table 4: Parameter-level L0, spectral L0, and absolute SV magnitudes
BF16 has high rank@90% because the surviving nonzero entries are randomly scattered (no low-rank structure),
but the absolute SV magnitudes ($\|\Delta W\|_F$, $\Sigma \sigma_i$) are 5-20 $\times$ smaller — the "high rank" captures almost nothing

Optimizer	Obj.	Prec.	% Params Changed	$\ \Delta W\ _F$	Rank @90%	% SVs < 1e-5	$\Sigma \sigma_i$ (nuclear)	mean(σ_i)
AdamW	GRPO	FP32	100.0%	0.0327	190	1.8%	0.6399	5.04e-04
AdamW	GRPO	BF16	2.1%	0.0063	732	2.4%	0.2156	1.65e-04
AdamW	GRPO	Mixed	100.0%	0.0334	329	1.3%	0.8069	6.31e-04
Muon	GRPO	FP32	99.9%	0.0214	752	1.3%	0.7469	5.78e-04
Muon	GRPO	BF16	0.5%	0.0014	590	19.2%	0.0435	3.63e-05
Muon	GRPO	Mixed	99.9%	0.0209	748	1.3%	0.7271	5.66e-04
AdamW	SFT	FP32	100.0%	0.0387	450	1.1%	1.0558	8.16e-04
AdamW	SFT	BF16	2.1%	0.0187	717	0.9%	0.6367	4.90e-04
AdamW	SFT	Mixed	100.0%	0.0467	379	0.9%	1.1533	8.96e-04
Muon	SFT	FP32	99.9%	0.0333	794	0.8%	1.1756	9.09e-04
Muon	SFT	BF16	0.4%	0.0019	492	19.7%	0.0543	4.75e-05
Muon	SFT	Mixed	99.9%	0.0321	730	1.0%	1.0981	8.66e-04